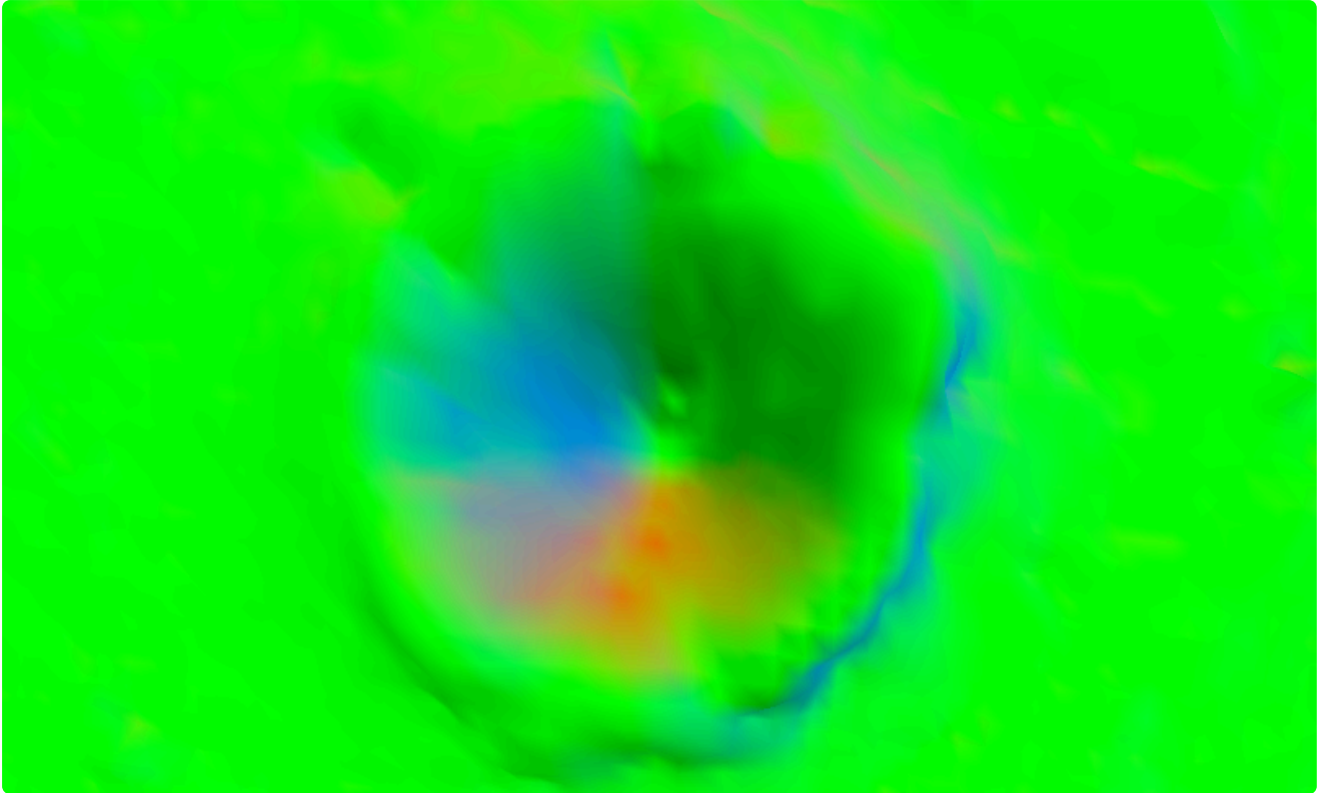


Assignment 3 TDT4195 Fall26

Task 1

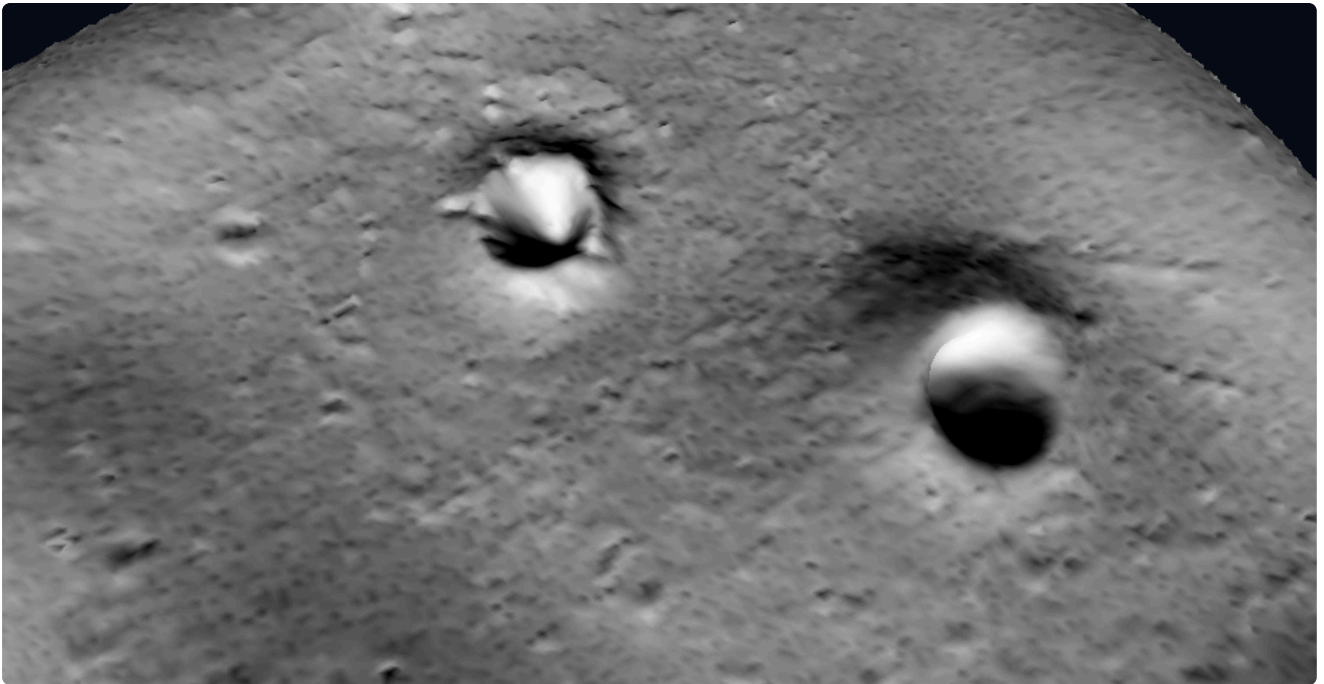
c.)



Updated the color in my fragment shader using the norms.

```
void main()
{
    fragColor = vec4(norms, 1.0);
}
```

d.)



Updated my fragment shader to use the Lambertian model and left alpha alone to prevent it from becoming transparent if lighting is 0.

```
void main()
{
    vec3 lightDirection = normalize(vec3(0.8, -0.5, 0.6));
    vec3 norms_new = normalize(norms);
    float diffuse = max(0.0, dot(norms_new, -lightDirection));
    fragColor = vec4(color.rgb * diffuse, color.a);
}
```

Task 2

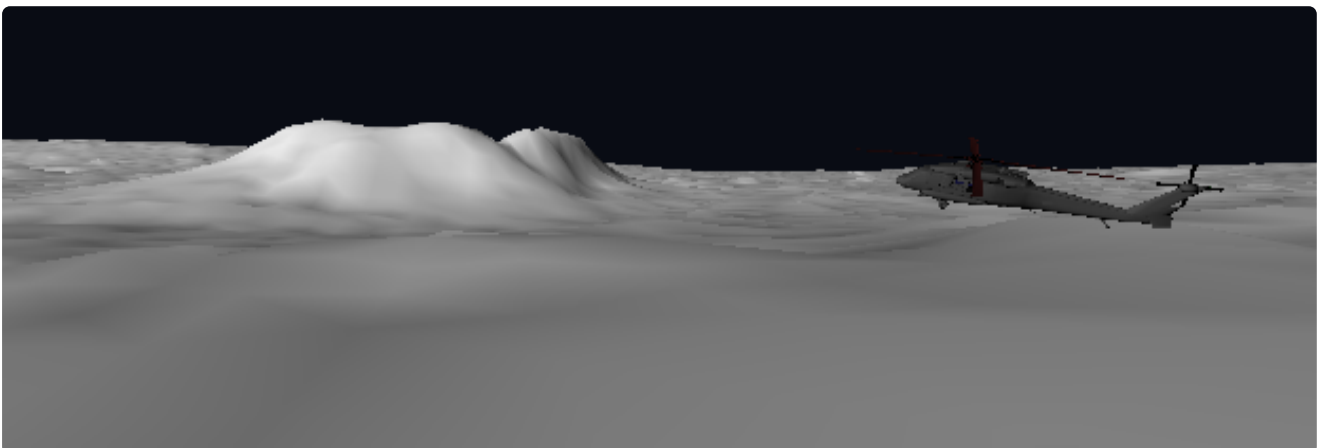
c.)

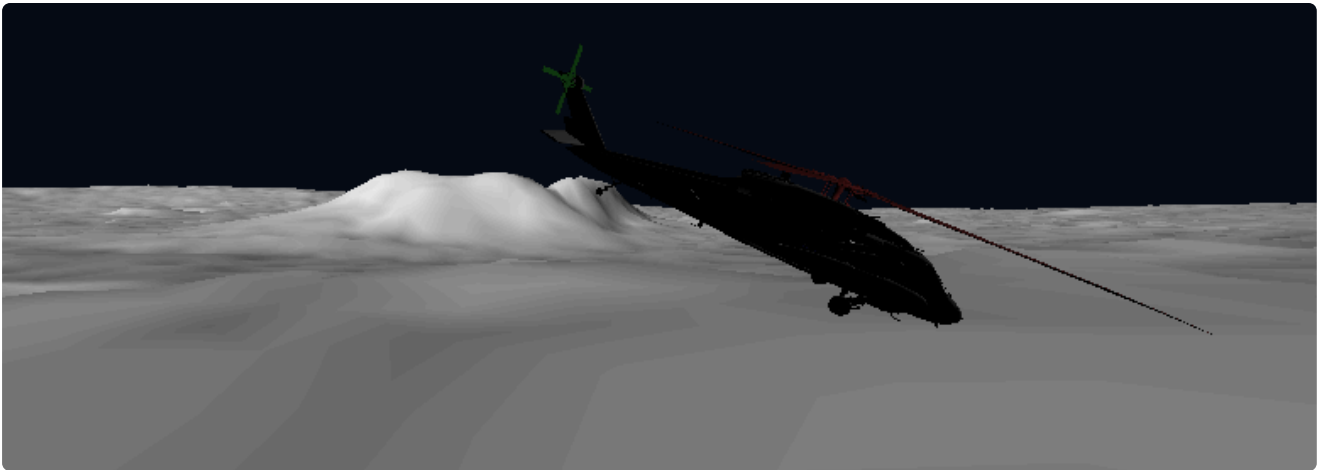


Task 5

a.)

One side turns "black" while the other remains normal.





c.)

Updated the vertex shader to recalculate the normals using the model matrix:

```
void main()
{
    gl_Position = matrix * vec4(position, 1.0);
    color = aColor;
    norms = normalize(mat3(model_matrix) * aNorms);
}
```

Now both sides are the same, showing the mesh and not just black.

